



Regulation of vitellogenesis by selected endocrine modulators in crab *Oziothelphusa senex senex*, with special reference to methyl farnesoate

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ABSTRACT

Using vitellogenin (Vtg) gene as a biomarker, the possible role of selected endocrine modulators (EMs) such as methyl farnesoate (MF), 17 α -hydroxy progesterone (17 α -OHP), biochanin-A, methionine enkephaline (ME) and arachidonic acid on female reproduction in crab *Oziothelphusa senex senex* (*O. senex senex*) was elucidated in this study. The injections were given on 1st and 7th day of experiment (10⁻⁸ moles/crab) of each EMs to its respective groups and corresponding concurrent controls (CC) received solvents of EMs and sacrificed on 14th day of experiment. Found hepatopancreatic Vtg expression in MF and 17 α -OHP injected crabs with significant ($p < 0.05$) increase in ovarian index (OI), oocyte diameter (OD) and ovarian Vtg levels and this was supported by RT-PCR results. Densitometry result of RT-PCR says that the hepatopancreatic Vtg expression is high in MF (74.51 ng) injected group than 17 α -OHP (48.62 ng) injected group. Due to this, further group of females were injected with 10⁻⁸ moles/crab of MF in an alternative day along with the groups of eyestalk ablated (ESX) and CC for 28 days. Significant reduction in length of reproductive cycle (13 ± 2 days) by MF injections than ESX and CC, along with significant ($p < 0.05$) increase in OI, OD and ovarian Vtg levels were observed. In *in vitro*, the isolated hepatopancreas fragments were incubated in M199 medium with and without MF (at 10⁻⁸ moles/well) at 25 ± 1 °C in CO₂ incubator. Vtg expression in hepatopancreas was found at 3rd and 6th hour of incubation *in vitro* and observed high density bands at 3rd h (67.21 ng) than 6th hour (56.90 ng) in RT-PCR. This is the first report stating the direct effect of MF on hepatopancreatic Vtg gene expression in crab *O. senex senex*. The results clearly suggest that MF and 17 α -OHP have stimulatory effect on vitellogenesis and MF has direct effect on crab *O. senex senex* hepatopancreatic Vtg expression.

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1. Introduction

Reproduction is one of the chief physiological events in the life cycle of an organism. In crustaceans vitellogenesis symbolizes the reproduction and is mediated by complex hormonal coordination which has become a fascinating area of research. Significant

research has been done on the vitellogenesis, aid in the nourishment of the developing embryo. Usually, vitellogenesis takes place in fat bodies (insects) and hepatopancreas and/or ovary (Raikhel and Dhadialla, 1992; Avvarre et al., 2003; Girish et al., 2014). Whereas regulatory factors of vitellogenesis were identified in the brain, thoracic ganglion, ovary along with eyestalk X-organ sinus gland complex, mandibular organ (MO) and Y-organ (Y-O).

ESX is a classical method employed to induce the reproduction and molting in crustacean aquaculture industry (Chaves, 2000; Rotllant et al., 2000). Eyestalk is involved in synthesis and secretion of an array of hormones including vitellogenesis inhibiting hormone (VIH) and MOIH which inhibits ovarian maturation (Swetha et al., 2011). The ESX enforce the animal enter in to reproduction. But the cautery of eyestalk leads to loss of hemolymph which ultimately causes death in the brood stock and inferior quality seed production. Therefore the scientists are in search of alternative, non-surgical methods to induce reproduction in crustaceans.

Abbreviations: Vtg, vitellogenin; EMs, endocrine modulators; MF, methyl farnesoate; 17 α -OHP, 17 α -hydroxyprogesterone; CC, concurrent control; ESX, eyestalk ablated; OI, ovarian index; OD, oocyte diameter; MO, mandibular organ; Y-O, Y-organ; MOIH, mandibular organ inhibiting hormone; ME, methionine enkephalin; BC-A, biochanin-A; PBS, phosphate buffer saline; DMSO, di-methyl sulphoxide; OvsVtg, *Oziothelphusa senex senex* vitellogenin; RT-PCR, reverse transcriptase-polymerase chain reaction.

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MF a sesqui terpenoid, unepoxidated analog of juvenile hormone—III of insects, released from MO is nominated for its involvement in regulating reproduction (Wainwright et al., 1996; Borst et al., 2001; Reddy and Reddy, 2006). MF acts as a positive regulator of reproduction in crustaceans. The release of MF is under the control of eyestalk neuro peptide hormone mandibular organ inhibiting hormone (MOIH). Though the MF induced ovarian maturation has been reported in many crustaceans the studies are restricted only in witnessing the morphological differences of ovarian indices (OI and OD), but the mode of action of MF has not been revealed yet.

In addition to MF, several lines of evidence suggest the involvement of many other EMs that direct the process of ovarian maturation viz., 17α -OHP (Reddy et al., 2006) ME (Sarojini et al., 1995, 1997; Kishori and Reddy, 2005) 17β -estradiol (Coccia et al., 2010) dopamine (Prasad et al., 2014) and other amines. Reports on the role of these EMs are still in its infancy and in depth studies are required in this direction. An attempt is made to understand the role of selected endocrine modulators of different nature such as MF (terpenoid), 17α -OHP (steroid), arachidonic acid (estrogen), ME (opioid) and BC-A (isoflavanoid) on Vtg expression levels in hepatopancreas of crab. Further special emphasis was laid by *in vivo* and *in vitro* studies on MF regulated reproduction (*in vivo*) and mode of action on hepatopancreatic Vtg expression (*in vitro*) in crab *Oziothelphusa senex senex*.

2. Materials and methods

2.1. Maintenance of animals

Freshwater intact female *O. senex senex* were obtained from rice fields in and around Renigunta and Tirupati (13.6°N, 79.5°E), A.P, India. Crabs were maintained in laboratory in large tubs (3–4 crabs/tub) at room temperature $27 \pm 1^\circ\text{C}$ with controlled light (12 h). The crabs were fed with sheep meat *ad libitum* in an alternate days and water was replaced daily.

2.2. Endocrine modulators (EMs)

The selected EMs for the present study are MF (Echelon Biosciences, Salt Lake City, USA), arachidonic acid, BC-A, methionine enkephaline (Sigma–Aldrich St. Louis, MO, USA) and 17α -OHP (German Remedies, from local drug house). MF, 17α -OHP and arachidonic acid were dissolved in absolute ethanol, BC-A in DMSO and methionine enkephaline in PBS up to the required concentration of 10^{-8} moles.

2.3. Effect of selected EMs on Vtg expression in crab

Total of 45 crabs were divided in to 9 groups of each 5. First group served as initial control and sacrificed on day 0 of the experiment. Groups 2–4 served as CCs and received 95% ethanol, PBS and DMSO volume equivalent to their respective EMs respectively. Groups 5–9 received 10^{-8} moles/crab of MF, arachidonic acid, 17α -OHP, ME and BC-A respectively on 1st and 7th day of experiment. All the CCs and the test group animals were sacrificed on 14th day of the experiment. The OI, OD and ovarian Vtg levels were measured and the hepatopancreas was isolated, cleaned and used for total RNA isolation.

2.4. Measurement of OI, OD and ovarian Vtg levels

OI and OD were estimated by Reddy et al. (2006) and Vtg from ovaries of control and experimental groups were isolated by using the method described by Tsukimura et al. (2000). The quality of

isolated Vtg was determined by SDS-PAGE and stored at -20°C for further estimation.

The isolated Vtg was estimated by Enzyme Linked Immune Sorbent Assay (ELISA). The well of certified 96-well microtiter plate were coated with $20\mu\text{l}$ of concentrated sample diluted in 1:10 ratio with coating buffer (carbonate buffer: 1.59 g sodium carbonate, 2.93 g sodium bicarbonate in 1000 ml distilled water, diethyl dithio carbamate 56 mg per 25 ml buffer). The wells coated with buffer alone were treated as blank. The covered plate was incubated in humid chamber at 37°C for 2 h. After incubation contents were discarded and the plate was filled with $200\mu\text{l}$ of diluted (1:1000) primary antibody (antibodies raised in rabbits against Oss-Vtg isolated from vitellogenic stage III ovary) in 0.1 M phosphate buffer saline containing 0.05% tween 20, pH 7.2 (PBST) with 2% polyvinyl pyrrolidone, 0.2% ovalbumin (PBST-PO). Incubated for 2 h at 37°C followed by discarding, the plate was filled with $200\mu\text{l}$ of horseradish peroxidase (HRP) conjugated anti IgG antibody (1:1000 dilution with PBST-PO; purchased from Genei, Bangalore) and incubated for 1 h at 37°C . Each step is followed by successive washing of plate with PBST for 3 times. Finally the plate was added with $200\mu\text{l}$ tetra methyl benzidine (TMB in 0.015% hydrogen peroxide) and incubated for 1 h at 37°C . The reaction was stopped by adding $50\mu\text{l}$ of 1.0 M phosphoric acid per well. Absorbance of each well was measured with ELISA reader (Bio-Rad I mark, USA) at 450 nm. All standards and samples measurement were performed in duplicate.

2.5. Primer designing and PCR amplification

The gene specific primers for *O. senex senex* Vtg (OssVtg) and β -actin used in earlier study (Girish et al., 2014) were used for the present study. Total RNA from hepatopancreas of all the experimental groups were extracted by using GeneJET RNA Purification Kit (Thermo Scientific # K0731). By taking $1\mu\text{g}$ of total RNA, cDNA was synthesized using RevertAid First Strand cDNA synthesis kit (Thermo Scientific # K1691). Using gene specific primers of Vtg (VtgF: 3'CAGATATGTCCGCAACACA5' and VtgR: 3'GGATAAACCCACGAACCT5') and β -actin (β -ActinF: 3'CAAGCGAGGTATCCTGACTCT 5' and β -ActinR: 3'CCACGTTTCATCTCACTCTCG5') cDNA was subjected for PCR (Girish et al., 2014). The PCR amplified products were examined in 1.5% agarose gel.

2.6. Effect of MF on crab reproduction

2.6.1. In vivo effect of MF

For *in vivo* studies crabs were divided into 4 groups, including initial control (5 crabs), CC (20 crabs), ESX group (45 crabs) and MF injected group (40 crabs). The volume of hemolymph and injected volume of MF is calculated according to Reddy (1980) and Tamone and Chang (1993). On day 1 of experiment all initial control crabs were sacrificed. CCs were maintained along with other experimental groups by injecting equivalent volumes of 95% ethanol alone on par with MF injected group and the third group crabs were ablated eyestalk on day 1 of experiment. The MF injections were given every alternative day up to 28 days at a concentration of 10^{-8} moles/crab at the base of third walking leg to the 4th group. The animals in experimental groups (both MF injected and ESX) and the CCs were sacrificed on 7th, 14th, 21st and 28th day of experiment. On respective days of sacrifice the color of ovary, OI, OD and ovarian Vtg levels were measured as per the standard protocols.

2.6.2. In vitro effect of MF

The hepatopancreas from immature crabs were isolated, cleaned and each one made in to 4 fragments. The first fragment was incubated in the first well of culture plate with 1.5 ml of medium 199 and 0.5 ml of crustacean physiological saline, and considered as

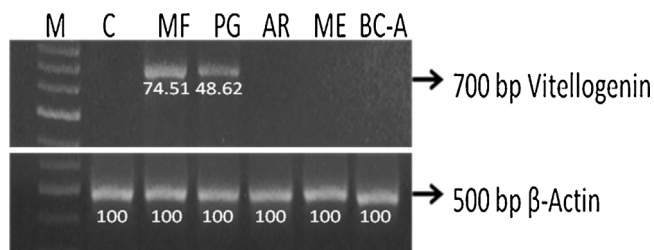


Fig. 1. The regulation of selected endocrine modulators on the transcription of hepatopancreatic vitellogenin gene in crab *Oziothelphusa senex senex*. The values beneath the bands indicate the concentration (ng) of the amplified product in RT-PCR.

M: marker, MF: methyl farnesoate, PG: 17 α -hydroxy progesterone, AR: arachidonic acid, ME: methionine enkephaline, BC-A: biochanin-A.

control. Other 3 fragments were incubated in 2nd, 3rd and 4th wells of the culture plate having 1.5 ml of medium 199 and 0.5 ml of crustacean saline with MF at a final concentration of 10^{-8} moles/well. The plate was incubated in CO₂ incubator at $25 \pm 1^\circ\text{C}$ in sterile conditions. Total RNA was isolated from each fragment (Thermo Scientific # K0731) at 3rd hour (1st and 2nd well), 6th hour (3rd well) and 9th hour (4th well) of incubation. After cDNA synthesis (Thermo Scientific # K1691) all the samples were subjected to PCR amplification with primers VtgF vs VtgR and β -ActinF vs β -ActinR. This experiment was done for 3 crabs and the PCR product (700 bps in size) of each fragment was examined in 1.5% agarose gel.

2.7. Densitometric analysis of RT-PCR products

The gel images of RT-PCR products were subjected to the software AlphaEaseFC Version 4.0 (Alpha Innotech Corporation, California, USA) to analyze and determine the concentration (ng) of RT-PCR product.

2.8. Statistical analysis

The data were analyzed statistically by performing Paired *t*-test and Two Way ANOVA followed by Bonferroni post test. Results were expressed as mean $\pm$ SEM. The experimental values at $p < 0.05$ was considered as significant.

3. Results and discussion

Among all the EMs received crabs, observed significant increase in OI ($p < 0.001$), OD ($p < 0.001$) and ovarian Vtg levels ($p < 0.001$) in MF and 17 α -OHP injected groups (Table 1). In RT-PCR studies of hepatopancreas observed Vtg expression only in MF (74.51 ng) and 17 α -OHP (48.62 ng) injected groups (Fig. 1). It is clear from the results that arachidonic acid, ME and BC-A does not have any role in the ovarian maturation of crab. The obtained results are in consistent to the existing publicational evidences of progression of ovarian maturation up on administration of MF in *Procambarus clarkii* (*P. clarkii*) (Laufer et al., 1998) and *O. senex senex* (Nagaraju et al., 2006) and MF in combination with 17 β -estradiol in *P. clarkii* (Rodriguez et al., 2002a). Besides this studies on stimulatory role of farnesoic acid the precursor of MF on hepatopancreatic Vtg expression in time dependent manner by *in vitro* in *Charybdis feriatus* (Mak et al., 2005), supporting the role of MF induced reproduction in crab in the present study.

Though studies claim the stimulatory role of 17 α -OHP in ovarian maturation of *O. senex senex* (Reddy et al., 2006) on the other hand it also carries negative impact on vitellogenesis when combined with MF (Rodriguez et al., 2002a). Convincing reports on the positive effect of estrogens (17 α -OHP and 17 β -estradiol) on vitellogenesis (Rodriguez et al., 2002b) and the correlating lev-

els of estrogens (17 α -OHP, 17 β -estradiol and progesterone) in the hemolymph with respect to vitellogenesis (Warrier et al., 2001; Zapata et al., 2003) are also available. In the present study the ovarian maturation in terms of increased OI, OD and ovarian Vtg levels has been relating to the expression of Vtg m-RNA in hepatopancreas of the crab *O. senex senex* by MF and 17 α -OHP. However the action of these two EMs may be either by direct epigenic action on hepatopancreatic Vtg expression or by indirect through stimulating other biological molecules (ecdysteroids from Y-O) thereby Vtg expression in crab hepatopancreas. Nevertheless the action of MF is higher in expressing the Vtg gene in hepatopancreas of crab than 17 α -OHP.

Though the high uptake of arachidonic acid by ovarian cells compared to other cells is evidenced (Reddy et al., 2004), no reports existing on the stimulatory effect of arachidonic acid on ovarian maturation and hepatopancreatic Vtg levels. Sufficient publicational reports on *in vivo* studies in *Uca pugnator* (Rotllant et al., 2000) and *Romalea microptera* (Kumar et al., 2012) and *in vitro* studies in *P. clarkii* (Sarojini et al., 1997) offers the involvement of the opioid ME in the suppression of ovarian maturation and observed results are in support to the result of present study. BC-A an estrogen mimic has found no sign of initiation of vitellogenesis, though the studies concerning to the occurrence of receptors of estrogen were identified in the hepatopancreas of *Austropotamobius pallipes* (Paolucci et al., 2002). It is proved from the present study that arachidonic acid, ME and BC-A has no role on reproduction or they may have negative effect on reproduction in crab.

With the positive (stimulatory) output of MF on vitellogenesis on ovarian maturation and Vtg gene expression (RT-PCR result; Fig. 1) studies than 17 α -OHP, further emphasis was laid in regard to MF as it is a natural hormone of crustaceans. *In vivo* studies were carried to illuminate the efficacy of MF on ovarian maturation, ovarian Vtg levels and length of induced reproductive cycle in crab *O. senex senex*. Two successive reproductive cycles were examined for validation with alternative days of administration of MF for about 4 weeks along with positive control as ESX group. From each group specified number of crabs was sacrificed on 7th, 14th, 21st and 28th day of experimentation. Found that all the CCs sacrificed during experimentation are in immature stage (5 crabs on each day). On 7th day (number of crabs i.e., $n = 4$) observed that the ovary of MF received animals is in Vtg stage III, on the other hand on 14th, 21st and 28th day MF received crabs were observed in Vtg stage II (14th day, $n = 5$; 21st day, $n = 4$; 28th day, $n = 5$) and III (14th day, $n = 5$; 21st day, $n = 4$; 28th day, $n = 6$). Where as in ESX group 7th–28th day of experimentation observed progression in the ovarian maturation i.e., Vtg stage I (7th day $n = 5$; 14th day $n = 5$), II (14th day $n = 5$; 21st day $n = 5$; 28th day $n = 5$) and III (21st day $n = 5$; 28th day $n = 6$). Ovulation was observed only in the MF injected group between 11 and 15 days ($n = 25$) and 26–28 days ($n = 7$) of experimentation. Crabs ovulated between 11 and 15 days were continued with MF injections after removing eggs from brood sac. During experimentation observed no mortality in control and MF injected crabs, except in ESX group (20%).

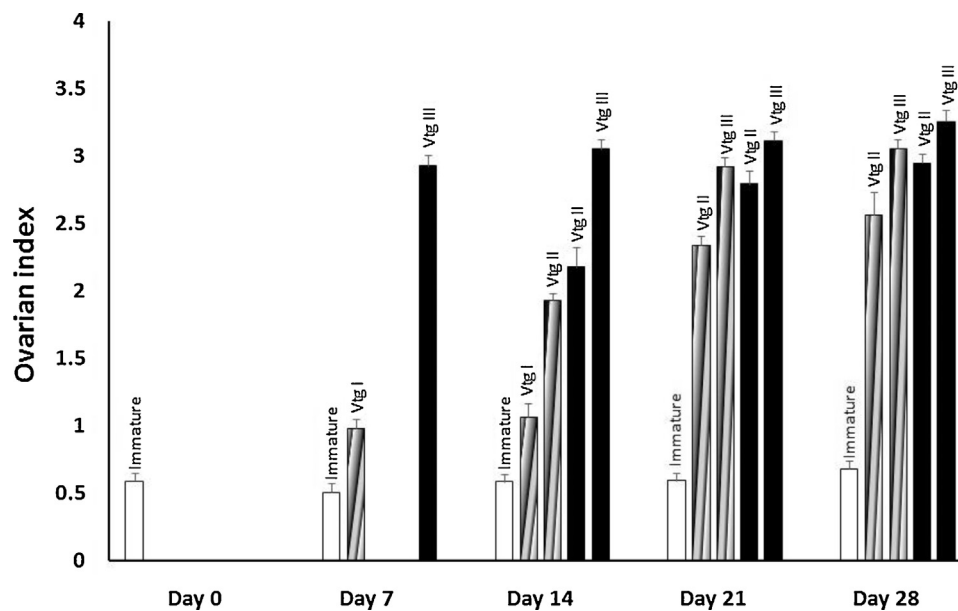
The OI, OD and ovarian Vtg levels were significantly ($p < 0.001$) increased in MF injected and ESX crabs and are depicted in Figs. 2–4 respectively in the *in vivo* experiment. It is observed that the mean value of OI for MF injected group (Vtg stage III: 2.93 ± 0.085) is significantly high when compared to CCs (Immature: 0.511 ± 0.069) and ESX group (Vtg stage I: 0.984 ± 0.069) on 7th day of experiment. On 14th day the mean values of OI in MF injected (Vtg stage II: 2.181 ± 0.149 ; Vtg stage III: 3.057 ± 0.073) and ESX (Vtg stage I: 1.067 ± 0.106 ; Vtg stage II: 1.937 ± 0.052) groups was significant when compared to CCs (Immature: 0.59 ± 0.056). Similarly on 21st and 28th day of experimentation observed significant increase in OI in the MF injected (Vtg stage II: 2.798 ± 0.098 and 2.954 ± 0.068 ; Vtg stage III: 3.12 ± 0.063 and 3.259 ± 0.085 respectively) and ESX

Table 1Effect of selected endocrine modulators on ovarian index, oocyte diameter and ovarian vitellogenin levels in crab *Oziothelphusa senex senex*.

Group	Ovarian index	Oocyte diameter (μm)	Vtg levels in ovary
Control	0.512 ± 0.069	209 ± 19.65	0.093 ± 0.008
Ethanol injected	0.589 ± 0.057 (15.03)	205 ± 11.73 (−1.91)	0.111 ± 0.015 (19.3)
MF injected	2.185 ± 0.189 (117.9) $p = 0.0005$	768 ± 34.70 (274.6) $p < 0.0001$	0.614 ± 0.021 (453.1) $p < 0.0001$
Arachidonic acid injected	0.577 ± 0.064 $p = 0.507$ (−2.03)	228 ± 8.6 $p = 0.0025$ (11.21)	0.082 ± 0.007 $p = 0.0426$ (−26.1)
17 α -OHP injected	1.068 ± 0.107 $p = 0.0012$ (81.3)	531 ± 52.16 $p = 0.0014$ (159.02)	0.531 ± 0.039 $p = 0.0002$ (378.3)
F-value	109.18 $p < 0.0001$	184.46 $p < 0.0001$	293.19 $p < 0.0001$
PBS (1%) injected	0.598 ± 0.037	190.4 ± 16.05	0.095 ± 0.008
Met-enkephaline injected	0.685 ± 0.05 $p = 0.052$ (14.5)	175 ± 8.83 $p = 0.115$ (−8.08)	0.099 ± 0.008 $p = 0.094$ (4.21)
F-value	14.77 $p = 0.0021$	8.14 $p = 0.018$	6.83 $p = 0.0186$
DMSO (100%) injected	0.530 ± 0.069	194.4 ± 8.40	0.087 ± 0.008
BC-A injected	0.583 ± 0.067 $p = 0.003$ (10)	214.8 ± 8.194 $p = 0.641$ (10.4)	0.108 ± 0.007 $p = 0.041$ (24.1)
F-value	7.75 $p = 0.0135$	2.5 $p = 0.1434$	8.14 $p = 0.0118$

The values are mean $\pm$ SEM of five individuals. Values in the parenthesis are percent change from respective solvent vehicle injected crabs. Values are significantly differ from controls at $p < 0.05$.

MF—methyl Farnesoate; 17 α -OHP—17 α hydroxyl progesterone; PBS—phosphate buffer saline; Met—enkephaline-methionine enkephaline; DMSO—dimethyl sulphoxide; BC-A—biochanin-A.

**Fig. 2.** Ovarian index in the crab *Oziothelphusa senex senex* at different time points after ESX and MF injections.

Vertical bars represent mean $\pm$ SEM.

Black color filled bars-MF injected; striped bars-ESX crabs; open bars-control and concurrent control crabs.

ESX-eye stalk ablated, MF-methyl farnesoate, Vtg-I-III represent vitellogenic stage I-III respectively.

groups (Vtg stage II— 2.341 ± 0.068 and 2.568 ± 0.167 ; Vtg stage III— 2.926 ± 0.067 and 3.064 ± 0.065 respectively) than CCs (Immature: 0.598 ± 0.057 ; 0.686 ± 0.0654 respectively) (Fig. 2).

Like OI, increase in the mean values of OD were found significant ($p < 0.001$) than CCs (Immature—7th day: 300.4 ± 33.35 ; 14th day:

309 ± 20.49 ; 21st day: 316.2 ± 40.46 and 28th day: 321.6 ± 39.21) in MF Injected and ESX crabs on 7th (MF—Vtg stage III: 2058 ± 45.01 and ESX—Vtg stage I: 762.8 ± 31.87), 14th (MF—Vtg stage II: 1331 ± 24.14 ; Vtg stage III: 2101 ± 52.26 and ESX—Vtg stage I: 788.4 ± 32.76 ; Vtg stage II: 1226 ± 55.03), 21st (MF—Vtg stage

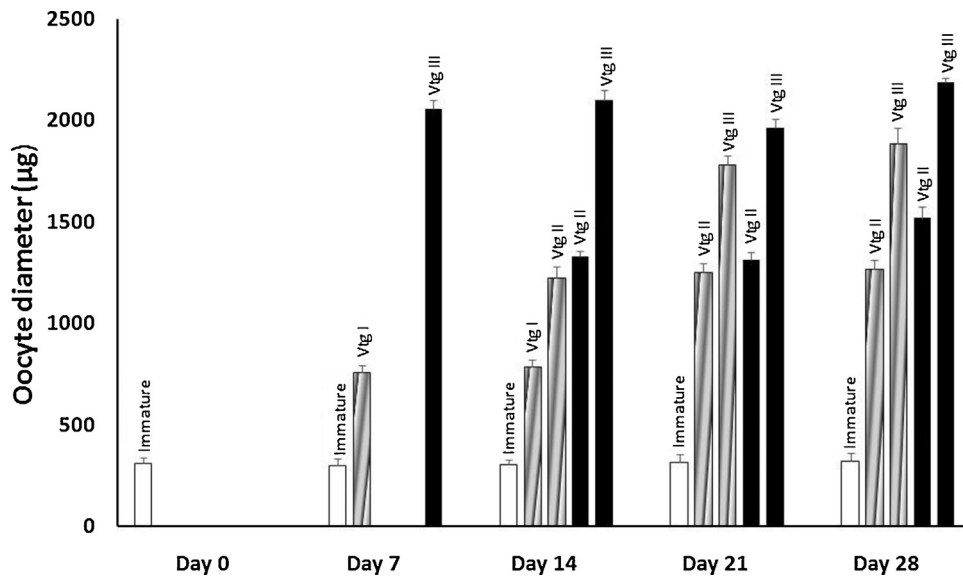


Fig. 3. Oocyte diameter in the crab *Oziothelphusa senex senex* at different time points after ESX and MF injections. Vertical bars represent mean $\pm$ SEM. Black color filled bars-MF injected; striped bars-ESX crabs; open bars-control and concurrent control crabs. ESX-eye stalk ablated, MF-methyl farnesoate, Vtg-I-III represent vitellogenic stage I-III respectively.

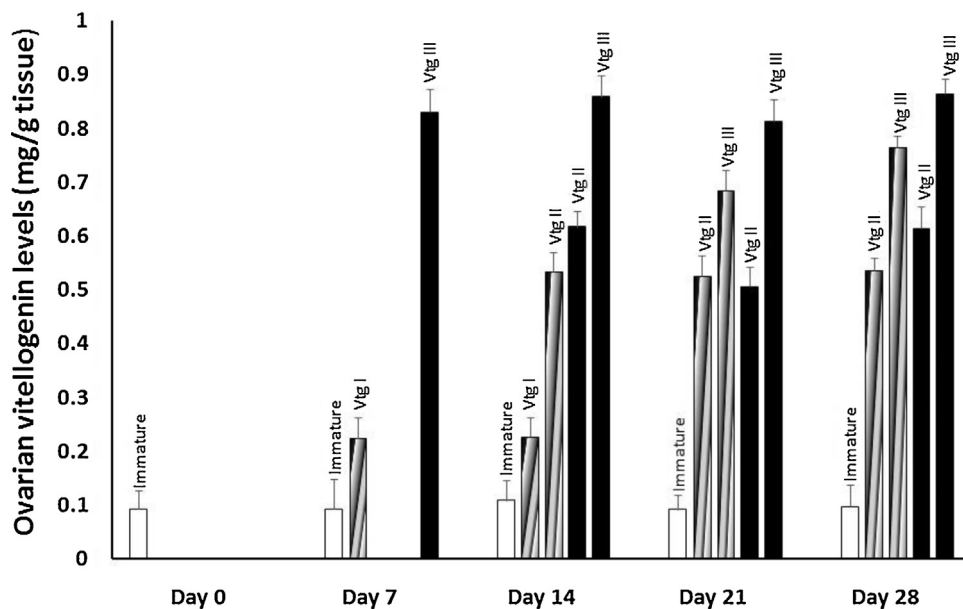


Fig. 4. Ovarian Vitellogenin levels in the crab *Oziothelphusa senex senex* at different time points after ESX and MF injections. Vertical bars represent mean $\pm$ SEM. Black color filled bars-MF injected; striped bars-ESX crabs; open bars-control and concurrent control crabs. ESX-eye stalk ablated, MF-methyl farnesoate, Vtg-I-III represent vitellogenic stage I-III respectively.

II: 1315 ± 36.3 ; Vtg stage III: 1966 ± 45.6 and ESX-Vtg stage II: 1256 ± 41.62 ; Vtg stage III: 1782 ± 44.61 and 28th (MF-Vtg stage II: 1519 ± 55.12 ; Vtg stage III: 2190 ± 23.5 and ESX-Vtg stage II: 1269 ± 42.42 ; Vtg stage III: 1887 ± 79.99) day of experiment (Fig. 3).

Significant increase ($p < 0.001$) in mean values of ovarian Vtg levels were observed in MF injected (7th day-Vtg stage III: 0.831 ± 0.044 ; 14th day-Vtg stage II: 0.618 ± 0.029 ; Vtg stage III: 0.859 ± 0.04 ; 21st day-Vtg stage II: 0.507 ± 0.036 ; Vtg stage III: 0.815 ± 0.04 and 28th day-Vtg stage II: 0.615 ± 0.04 ; Vtg stage III: 0.865 ± 0.029) and ESX (7th day-Vtg stage I: 0.226 ± 0.038 ; 14th day-Vtg stage I: 0.227 ± 0.038 ; Vtg stage II: 0.535 ± 0.036 ; 21st day-Vtg stage II: 0.527 ± 0.037 ; Vtg stage III: 0.685 ± 0.038 and 28th day-Vtg stage II: 0.538 ± 0.023 ; Vtg stage III: 0.765 ± 0.022)

crabs when compared to concurrent control group (Immature-7th day: 0.093 ± 0.056 ; 14th day: 0.11 ± 0.037 ; 21st day: 0.092 ± 0.027 ; 28th day: 0.098 ± 0.039) during the experimentation (Fig. 4).

ESX and MF induced reproduction was reported in many crustaceans. In crab *O. senex senex* ovarian maturation was reported in ESX and MF injected groups, but the MF injections were given with 1 week gap up to 4 weeks (Nagaraju et al., 2006). In *P. clarkii* (Tsukimura et al., 2000) and *Macrobrachium rosenbergii* (Okumura and Aida, 2001) reported the induction of ovarian maturation by MF administration and ESX. In the present study we observed two induced reproductive cycles by MF in 28 days. From the results it is clear that MF administration is effective in reducing the period of reproductive cycle as evidenced by the progression of OI, OD and

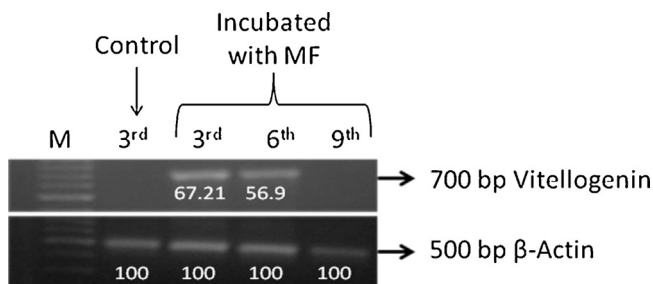


Fig. 5. The regulation of MF on the transcription of hepatopancreatic vitellogenin gene *in vitro* in crab *Oziothelphusa senex senex*. The values beneath the bands indicate the concentration (ng) of the amplified product in RT-PCR.

M: molecular weight marker; 3rd, 6th and 9th are the different time points (hours) of incubation.

ovarian Vtg levels over ESX in bringing vitellogenesis with the mean period of ovulation 13 ± 2 days. It is also found that MF injected animals were reached to Vtg stage III in a short period *i.e.*, in between 11 and 14 days, whereas the ESX group executed the same on 21–28 days. However the ovulation has not been recorded in ESX group which was identified in MF injected group. It is clear that MF due to its natural activity and direct action on vitellogenesis is able to induce reproduction in crab.

Further *in vitro* studies were conducted by incubating the hepatopancreas, with and without MF in medium 199 to disclose the mode of action of MF on vitellogenesis in crab. Observed the RT-PCR expression of Vtg gene in hepatopancreas incubated with MF at 3rd and 6th h of incubation with a product concentration of 67.21 and 56.9 ng respectively, but no expression has been observed at 9th h (Fig. 5). The only report available on Vtg expression in hepatopancreas is in *C. feriatus* (Mak et al., 2005) *in vitro*. The possible action of MF in inducing Vtg expression is supposed to take part through binding to the Retinoid X Receptor which is identified in ovary and hepatopancreas of crustaceans (Kim et al., 2005; Nagaraju et al., 2011). In similar to the existing hypothesis the MF incubated tissue showed sign of vitellogenesis initiation in early hour of incubation *i.e.*, 3rd hour and was carried up to 6th hour and as the time prolonged further the activity was found to be diminished. The initiation of vitellogenesis in the 3rd hour MF incubated sample and absence in the control sample at 3rd hour, clearly displays the direct stimulatory effect of MF on hepatopancreatic Vtg m-RNA expression thereby reproduction in *O. senex senex*. However the *in vitro* experiment clearly says that the MF has direct epigenic activity on Oss-Vtg expression in hepatopancreas.

4. Conclusion

From the present study it is concluded that MF and 17α -OHP having the direct or indirect action on ovarian maturation in crab *O. senex senex* by stimulating Vtg gene expression in hepatopancreas the only site for Vtg synthesis in crab. MF induced ovarian maturation is faster than ESX and shortens the length of reproductive cycle in crab *in vivo*. By the *in vitro* experiment it is also clear that MF has direct effect on Oss-Vtg expression in hepatopancreas, suggesting there may be epigenic activity of MF on Oss-Vtg gene which has to be further characterized. However the present study instigated to produce quality seed by using endocrine modulators in crustaceans. The quality of seed produced by MF and 17α -OHP induced maturation has to be quarantined further and the studies at molecular level are in progress in this laboratory to understand the MF induced vitellogenesis in the crab *O. senex senex*.

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